

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam
(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Date: 07-08-2023

Course Code	UCS2501	Course Name	Computer Networks					
Course Type	Theory	Course Category	Professional Core (PC)	L	T	P	E	C
				3	0	0	0	3
Regulation	2021		Academic Year & Batch	2023 - 24 & 2021 - 25				
Degree and Branch			B.E. Computer Science & Engineering					
Semester			V					
Department Offering the Course			Computer Science and Engineering					
Name of the Faculty			Dr. S. V. Jansi Rani & Dr. N. Sujaudeen					

OBJECTIVES

- To understand the protocol layering and physical level communication
- To understand the various components required to build different networks
- To learn the functions of network layer and the various routing protocols
- To familiarize the functions and protocols of the Transport layer
- To understand various application layer protocols.

UNIT I INTRODUCTION AND PHYSICAL LAYER

7

Introduction: Networks – Network types – Protocol layering – TCP/IP Protocol suite – OSI model;
Physical Layer: Performance; Socket Programming; Transmission media.

UNIT II DATA-LINK LAYER AND MEDIA ACCESS

9

Introduction: Link-Layer addressing – Error Correction and Detection: Checksum, Hamming Code –
DLC services – Data-Link layer Protocols – HDLC – Media Access Control – Wired LANs- Standard
Ethernet: Characteristics, Address, Access Method – IEEE 802.11 project– Connecting devices.

UNIT III NETWORK LAYER

11

Network layer services – Switching – Performance – IPV4 Addresses – Forwarding of IP packets;
Network Layer Protocols: IP – ICMP v4 – Unicast routing algorithms – Protocols – Multi- casting
basics – IPV6 addressing – IPV6 protocol.

UNIT IV TRANSPORT LAYER

9

Introduction: Transport layer protocols – Services – Port numbers – User datagram protocol –
Transmission control protocol.

UNIT V APPLICATION LAYER

9

Traditional applications – Electronic mail (SMTP, POP3, IMAP, MIME) – HTTP – File transfer
protocol – Secure shell (SSH) – DNS.

TOTAL PERIODS: 45

Course outcomes:

After the completion of this course, students will be able to:

CO 1: Describe the principles, design, terminology and concepts of the network models (K2)

CO 2: Select the protocols of data link layer and apply error control mechanisms (K3)

CO 3: Choose the core functions of network layer protocols and apply them for data communication (K3)

CO 4: Select the Transport Layer Protocols suitable for transmission of data. (K3)

CO 5: Illustrate the Application Layer Protocols and their basic functionalities (K2).

Content Delivery Methods (CDM): P- Presentations, D- Demo, PS –Problem Solving, In-Class Quiz (ICQ)

Delivery Tools: PowerPoint presentations\ Whiteboard applications\ Projector\ Board\ LMS

Assessment Tools: Assignment, Team Project

SNo	Topic	K level	CO	CDM	No. of Sessions		Deviations if any
					Proposed	Actual	
UNIT – I INTRODUCTION AND PHYSICAL LAYER (CO1)					7		
1	Introduction to Computer Networks and course overview	K2	CO1	P	1		
2	Introduction: Networks: Components, Data Representation, Data Flow, Network types: LAN, WAN	K2	CO1	P	1		
3	Protocol layering: Principles, logical connections, TCP/IP Protocol suite: Layered architecture, Layers and description, encapsulation and decapsulation	K2	CO1	P	1		
4	OSI model: Layers, OSI Vs TCP/IP	K2	CO1	P, ICQ	1		
5	Physical Layer: Performance: Bandwidth, Throughput, Latency, BWD, Jitter	K3	CO1	P, PS	1		
6	Socket Programming: System Calls, Client to Server connection	K3	CO1	P, Demo	1		
7	Transmission media: Guided Media- Twisted pair cable, Coaxial cable, Fiber-optic cable, Unguided Media: Radio Waves, Microwaves, Infrared	K2	CO1	P	1		
UNIT – II DATA-LINK LAYER AND MEDIA ACCESS (CO2)					9		
8	Introduction: Link-Layer addressing : Unicast, Multicast, Broadcast, ARP, RARP	K2	CO2	P, ICQ	1		
9	Error Correction and Detection: Checksum, Hamming Code	K3	CO2	P, PS	1		
10	DLC services: Framing, Flow and Error Control Data-Link layer Protocols: Simple Protocol, Stop and Wait protocol, Piggybacking	K3	CO2	P	1		
11	HDLC: Configurations, Transfer Modes	K2	CO2	P	1		
12	Media access control: Random Access, Controlled Access, Channelization	K3	CO2	P, PS	1		
13	Wired LANs: Standard, Fast, Gigabit Ethernet	K3	CO2	P	2		
14	IEEE 802.11: Architecture, MAC Sublayer, Addressing mechanism, physical layer	K2	CO2	P	1		

15	Connecting devices: Hubs, Switches, Routers	K2	CO2	P	1		
UNIT – III NETWORK LAYER (CO3)					11		
16	Network layer services: Packetizing, Routing and Forwarding	K2	CO3	P	1		
17	Switching: Circuit, Datagram and Message switching	K3	CO3	P, ICQ	1		
18	Network layer Performance: Delay, Throughput, Packet Loss	K3	CO3	P, PS	1		
19	IPv4 Addresses: Address space, classful and classless addressing, DHCP, NAT	K3	CO3	P, PS	2		
20	Forwarding of IP packets: Based on destination address, label	K2	CO3	P	1		
21	Internet Protocol (IP): Datagram Format, Fragmentation, Options	K2	CO3	P	1		
22	ICMP V4: Messages, Debugging Tools, ICMP Checksum	K2	CO3	P	1		
23	Unicast routing algorithms: Distance Vector – RIP, Link-State Routing – OSPF, Path-vector -BGP	K3	CO3	P	1		
24	Multicasting basics – Delivery at DL, Collecting information about groups, MC forwarding, Two Approaches	K3	CO3	P	1		
25	IPv6 addressing: Representation, Address space, Allocation, Autoconfiguration, Renumbering IPV6 protocol: Packet Format	K2	CO3	P	1		
UNIT – IV TRANSPORT LAYER (CO4)					9		
26	Introduction: Transport layer protocols: Simple, Stop and Wait, Go-Back-N Protocols, Services -- Port numbers	K2	CO4	P, ICQ	1		
27	User datagram protocol: User Datagram format, Services, Application	K3	CO4	P	1		
28	TCP Services, TCP Features	K2	CO4	P	1		
29	TCP Segment, TCP Connection	K3	CO4	P, PS	1		
30	TCP State Transition Diagram	K3	CO4	P	1		
31	Windows, Flow Control, Error Control	K2	CO4	P	1		
32	Congestion Control	K3	CO4	P, PS	2		
33	SCTP	K2	CO4	P	1		
UNIT – V APPLICATION LAYER (CO5)					9		
34	Application Overview	K2	CO5	P	1		
35	Electronic Mail SMTP : Architecture, Web based mail	K3	CO5	P, ICQ	1		
36	POP3	K2	CO5	P	1		
37	IMAP, MIME	K2	CO5	P	1		
38	HTTP	K3	CO5	P	1		
39	FTP: Two connections, Control and data connection	K3	CO5	P	1		
40	Secure Shell SSH : Components, Applications	K3	CO5	P	1		
41	Domain Name System DNS: Name Space, Resolution, Caching, Resource records	K3	CO5	P	1		
					45		
42	Content Beyond syllabus: Software Defined Networking: Why SDN, Components, Data and control plane, Architecture, Different models of SDN, Difference between SDN and traditional networking, Advantages and disadvantages of SDN	K2	CO5	P	1		
Total					46		

CO-PO/PSO mapping using Performance Indicators

PO/ PS O	CO1		CO2		CO3		CO4		CO5	
	PI	PI - M S	PI	PI - M S	PI	PI - M S	PI	PI - M S	PI	PI - M S
1	1.1.2 (3) 1.3.1 (3) 1.4.1 (3)	3	1.1.2 (3) 1.3.1 (3) 1.4.1 (3)	3	1.1.2 (3) 1.3.1 (3) 1.4.1 (3)	3	1.1.2 (3) 1.3.1 (3) 1.4.1 (3)	3	1.3.1(3) 1.4.1(3)	3
2	2.1.2 (3) 2.3.1 (3) 2.4.1 (2)	2	2.1.2 (3) 2.1.3 (2) 2.4.1 (2)	2	2.1.2 (3) 2.1.3 (2) 2.3.1 (3)	2	2.1.2 (3) 2.3.1 (3)	2	2.1.2 (3) 2.3.1 (3)	2
3	-	-	-	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-	-	-	-
5	-	-	-	-	-	-	-	-	-	-
6	-	-	-	-	-	-	-	-	-	-
7	-	-	-	-	-	-	-	-	-	-
8	-	-	-	-	-	-	-	-	-	-
9	-	-	-	-	-	-	-	-	-	-
10	-	-	-	-	-	-	-	-	-	-
11	-	-	-	-	-	-	-	-	-	-
12	-	-	-	-	-	-	-	-	12.1.2(2)	2
13	13.3.1(2)	2	13.3.1(2)	2	13.3.1(2)	2	13.3.1(2)	2	13.3.1(2)	2
14	-	-	-	-	-	-	-	-	-	-
15	-	-	-	-	-	-	-	-	-	-

CO to PO/PSO Mapping

	PO 1	PO2	PO 3	PO 4	PO5	PO 6	PO7	PO 8	PO 9	PO10	PO1 1	PO12	PSO 1	PSO2	PSO 3
CO1	3	2											2		
CO2	3	2											2		
CO3	3	2											2		
CO4	3	2											2		
CO5	3	2										2	2		
Total	15	10										2	10		
Score	3	2										2	2		

PI	Description	PI strength
1.1.2	Apply concepts of probability, statistics, and queuing theory in modelling of computer-based system and network protocols	3
1.3.1	Apply engineering fundamentals	3
1.4.1	Apply theory and principles of computer science and engineering to solve engineering problem	3
2.1.2	Identify processes/modules/algorithms of computer-based systems and parameters to solve problems	3
2.3.1	Apply computer engineering principles to build modules of a system with required applicability and performance	3
2.4.1	Analyze and interpret results using contemporary tools	2
12.1.2	Identify deficiencies or gaps in knowledge and demonstrate ability to find information to close these gaps	3
13.3.1	Apply fundamentals of problem solving	3

Assessment Tools for assessing Cos

Assessment Tool	CO1	CO2	CO3	CO4	CO5
CAT 1 (Units 1, 2, 3)	X	X	X		
CAT 2 (Units 3, 4,5)			X	X	X
Assignment 1 (K3) Units 1, 2	X	X			
Team Project (K5) Units 3, 4, 5			X	X	X
Semester Examination	X	X	X	X	X

Evaluation Criteria:

Description	Actual Marks	Weightage	Total Marks	Weightage
CAT 1	50	60%	200	40
Assignment 1	20	40%		
CAT 2	50	60%		
Team Project	20	40%		
Semester Examination	100	60%	100	60

Prepared By	Verified By	Approved By
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